

Introduction to the Data Pipeline and AI for CSOs

Lecture 1

Welcome

> About me

- > **Cedric Lombion** — Co-founder & CEO, Civic Literacy Initiative
- > 10+ years designing data capacity-building programs
- > Former Data & Innovation Lead, Open Knowledge Foundation (School of Data)
- > Advises World Bank, Transparency International, EITI
- > Adjunct lecturer at universities in Bordeaux and Turin



> About this course

- > **5 lectures** on using data, software, and AI for civic work
- > **5-day crash course** in April — hands-on project work in Turin or online
- > Designed for little to no programming experience
- > You will build a data project from scratch during the course
- > Today: how AI actually works, why it fails, and how the **data pipeline** keeps it useful

> About you

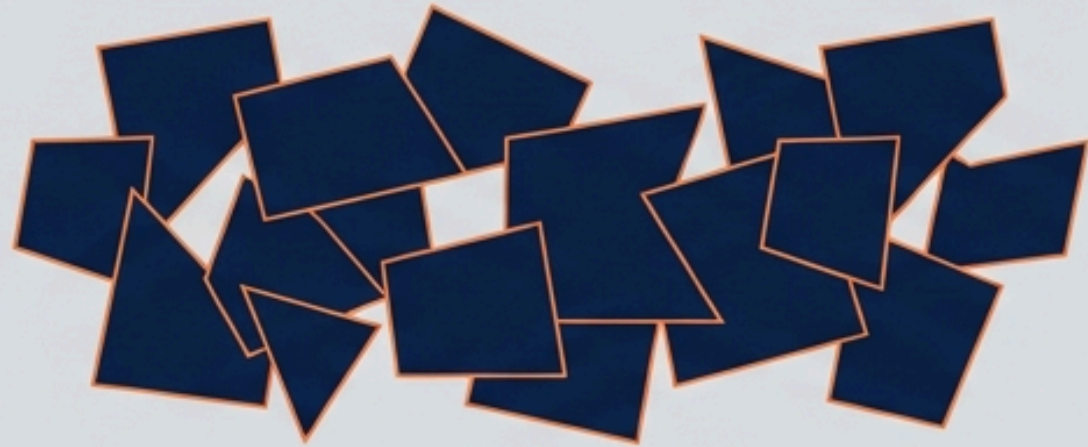
Let's find out with a quick quiz.



AI: How It Works, Why It Fails

> The civic data reality check

The Civic Data Reality



01

Contradictory Mandates

Local government standards conflict with donor reporting requirements.

02

Unintentional Hoarding

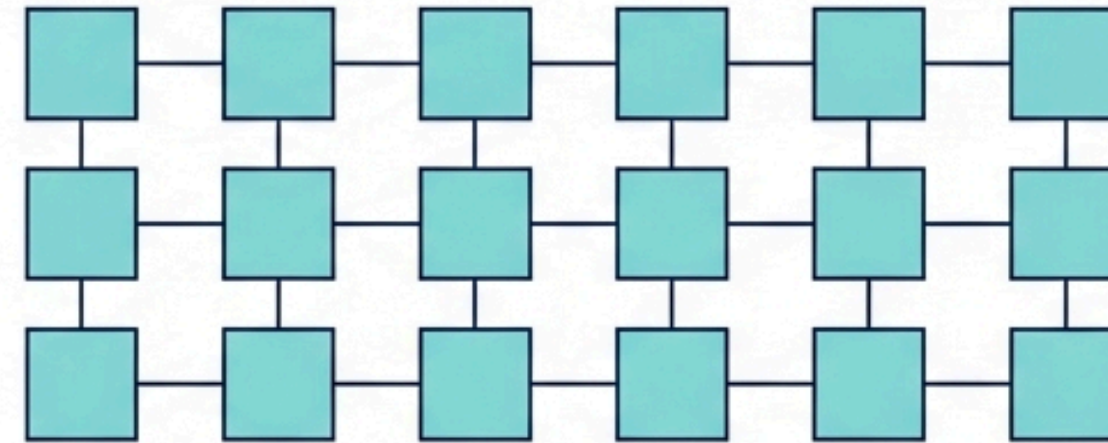
SMEs and NGOs collect vast amounts of data without clear, strategic intent.

03

Context-Heavy

Data is deeply tied to human realities, localized definitions, and complex biases.

The AI Assumption



01

Standardized Truth

AI assumes all inputs map to a single, universally agreed-upon factual reality.

02

Immediate Utility

AI assumes all stored numbers are clean, relevant, and ready for statistical prediction.

03

Context-Blind

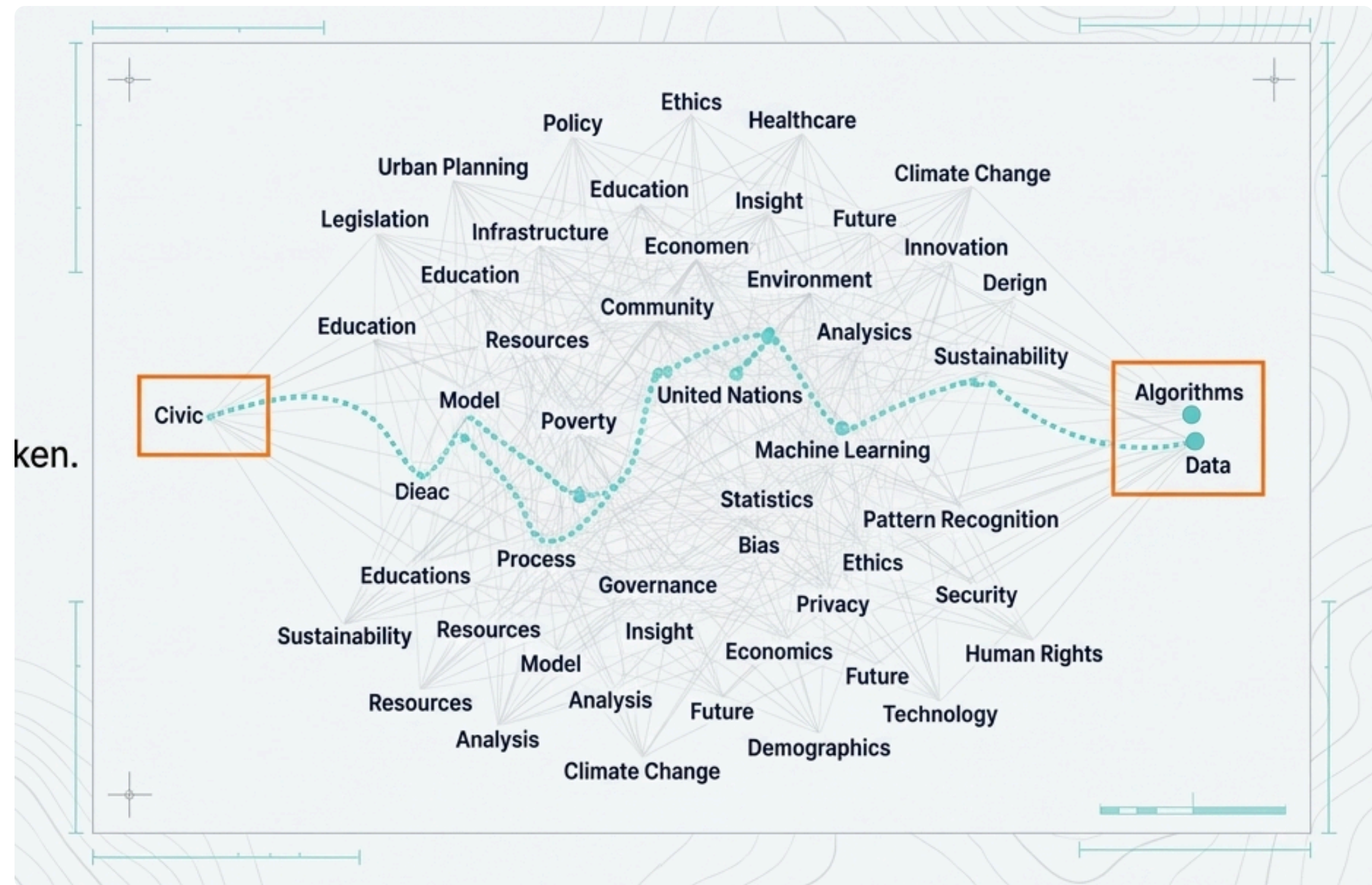
AI strips away localized nuance, optimizing only for the most statistically probable pattern.

> Navigating the probability cloud

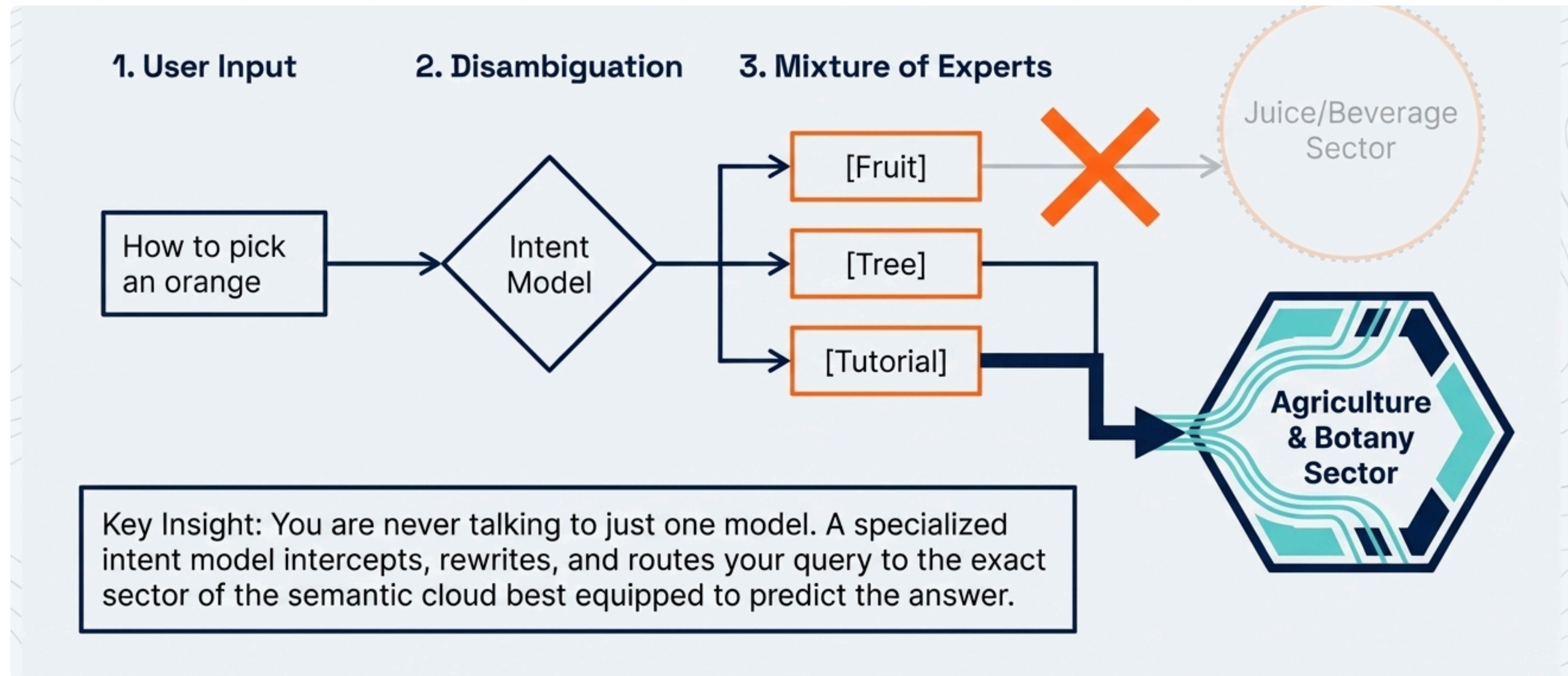
Large Language Models do not retrieve facts; they calculate probability of the next logical token.

When prompted, the AI navigates this space, connecting concepts mathematically.

It is a high-speed engine
connecting concepts, not a
database of knowledge.



> Query expansion and the expert router



> The intelligence illusion

	The Illusion	The Reality
Emotion	The AI feels anxiety or exhibits symptoms of depression.	The model's prediction path drifted into training data containing clinical studies on psychosis and depression. It simulates text, not feeling.
Reasoning	The AI understands the rules, strategy, and ultimate goal of a chess game.	It predicts the next plausible string of text in a series. After a few turns, it confidently makes illegal moves.
AGI Scale	Adding more data and compute will inevitably lead to an omniscient general intelligence.	Building a giant stairwell will not get you to the moon. Scaling statistical prediction does not spontaneously generate reasoning.

> Walking while looking at your feet

Paved Road

Well-represented data.
Predictions are reliable.

Narrow Road

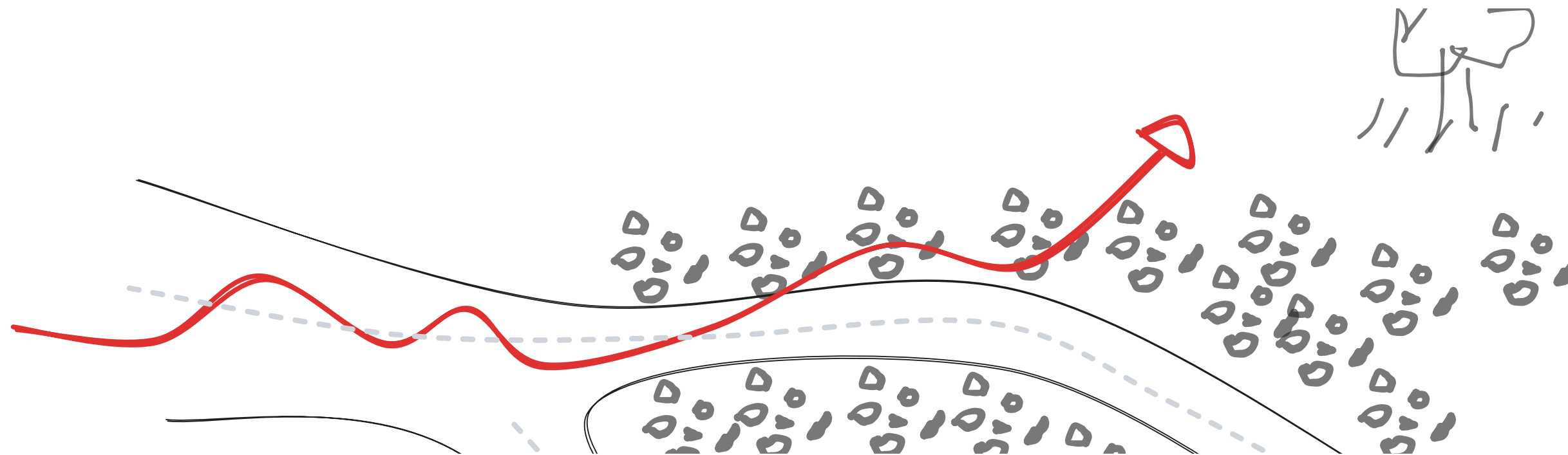
Niche civic data topics.
Less training data available.

The Gravel

Sparse, unreliable data.
The road narrows sharply.

Hallucination

No data at all. Pure
confabulation.

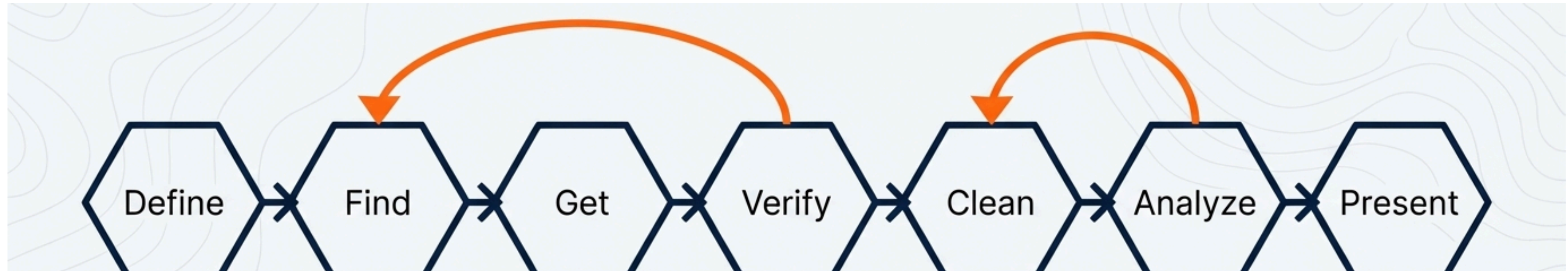


> From machine learning to generative AI

	Traditional Machine Learning	Generative AI Agents
Execution	Fixed algorithm processing strict, predefined inputs.	Flexible, step-by-step path adjustments based on rolling inputs.
Behavior	Explorable, highly tailored, and single-purpose.	Broadly capable (a "magic hammer"), but highly prone to hallucination.
Constraint	Bound entirely by underlying code architecture.	Requires explicit, external fences; capable of autonomous action including local file deletion.

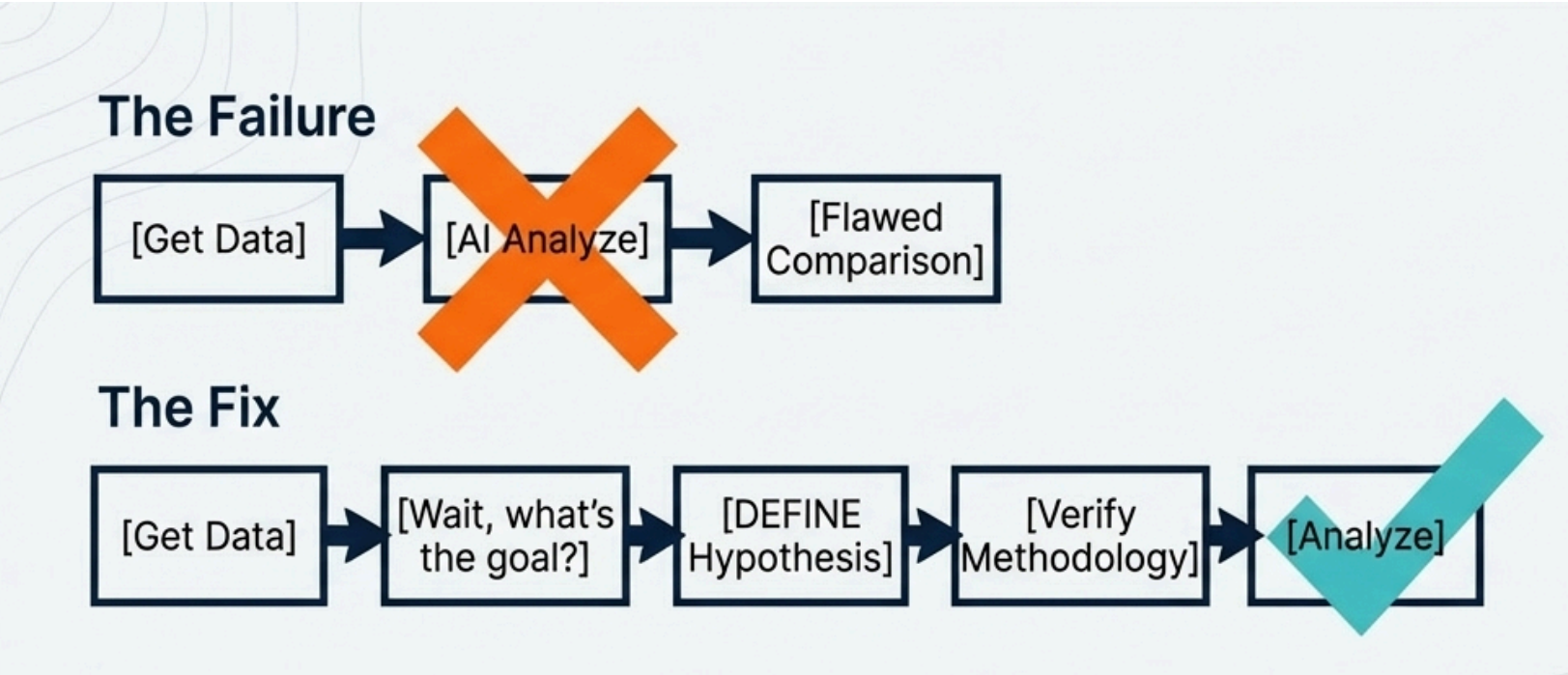
The Data Pipeline

> The pipeline as the ultimate harness



The best AI strategy is a rigorous traditional data strategy. Constrain AI to isolated steps — never let an agent jump from Get to Analyze without human verification.

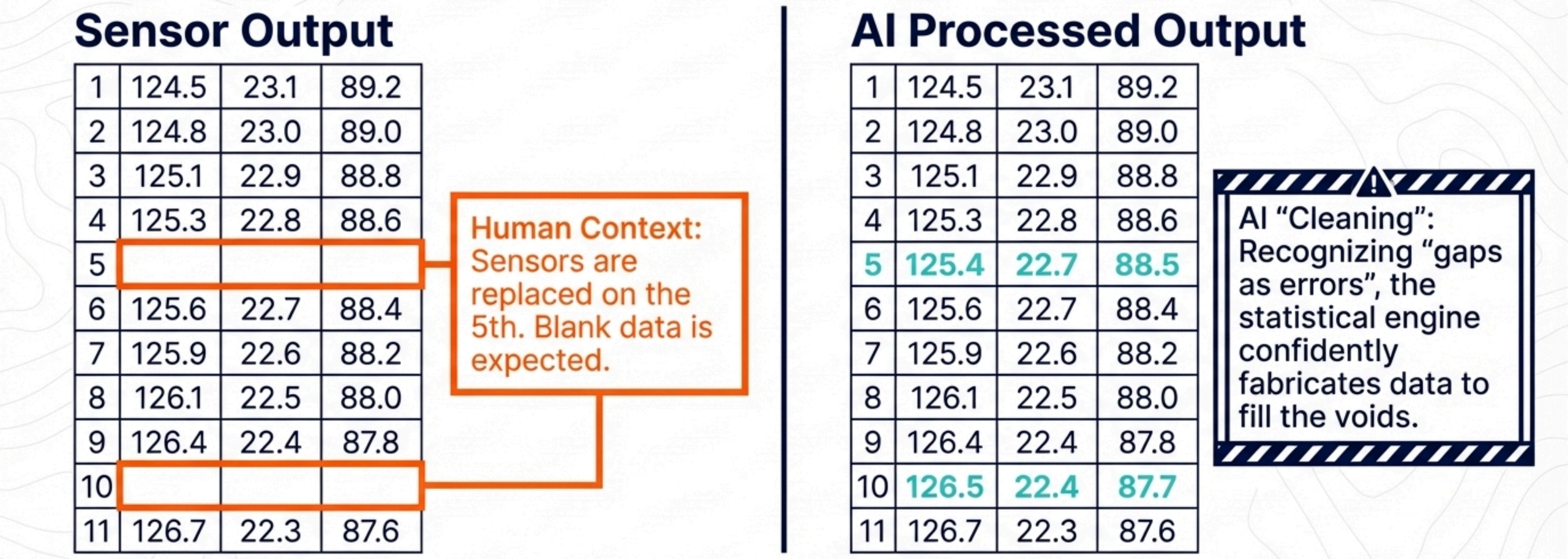
> The danger of the happy path



	Ivory Coast	Senegal
Year 1	1.2	0.8
Year 2	1.5	0.9

It's easy to jump straight to analysis. But without properly defining your goal and verifying the data, you compare data points that are not comparable — because they are defined differently by the institutions that produced them.

> The business logic blindspot



Generative AI understands the mathematical structure of a spreadsheet, but is entirely blind to organizational reality. Domain knowledge remains a strictly human responsibility.

> **Data management is a cross-cutting concern**

Beyond the 7 pipeline steps, every project needs:

- > **Storage** — where does the data live? Who can access it?
- > **Protection** — what's sensitive? What are the legal obligations?
- > **Anonymization** — can individuals be identified? Should they be?

These scale with project size — irrelevant for a Wikipedia exercise, critical for a national data collection.

The Define Step

> Define is where most projects fail

Not because people skip it — but because they think they've done it when they haven't.

Each one sharpens the next. A vague question produces a vague hypothesis, which produces a useless table.

> Define has three components

1. **Research question** — the specific question you want to answer
2. **Hypothesis** — your informed guess about what the answer might be
3. **Horizon table** — the empty spreadsheet that would let you test your hypothesis

> The research question defines the data

It is **specific**, **answerable with data**, and **relevant to a decision**.

Too vague

"How is homelessness in our region?"

Specific enough

"How does shelter funding per capita compare across 12 regions?"

"Is our programme working?"

"Did beneficiaries reaching employment increase after we changed intake criteria?"

If your question doesn't tell you **what data you need**, keep narrowing.

> Two starting points

From a dataset

"We have this data. What questions can we ask?"

Common in orgs that already collect data (M&E, surveys, case management).

The risk: you only ask questions your data can answer.

From a question

"We have this question. What data would answer it?"

Common in advocacy, investigations, policy analysis.

The risk: the data you need may not exist.

Both are valid. The important thing is to **articulate the question explicitly** — whichever direction you start from.

> A hypothesis makes your analysis honest

You declare what you expect *before* looking at the data.

- > It's not a guess — it's an **informed expectation** based on what you already know
- > It tells you what "interesting" looks like — and what would surprise you

A research question without a hypothesis leads to fishing for patterns. A hypothesis without a research question leads to confirmation bias.

> Design the empty spreadsheet before you look for data

The horizon table is the **output shape** — the exact table you'd need to test your hypothesis.

- > You design it *before* finding any data
- > It forces you to be specific: what is a row, what are the columns, what units, what sources

> How to build a Horizon Table

1. **Start with the question and hypothesis** What would you need to see in a spreadsheet to confirm or reject your hypothesis?
2. **Define the unit of observation** What is one row? A person? A region? A year? An event?
3. **Define the columns** What do you need to know about each unit to test the hypothesis?
4. **Add metadata columns** Source, date retrieved, confidence level.
5. **Name columns clearly** Lowercase, no spaces, descriptive: `funding_eur_per_capita`, not `Data2`.

The Define Step

A Concrete Example

> Shelter funding: question and hypothesis

A homelessness charity wants to make the case for redistribution of government shelter funding.

Research question: How does public shelter funding per capita compare to homelessness rates across the 12 regions of our country?

Hypothesis: Regions with higher homelessness rates receive proportionally *less* funding per capita — funding follows population, not need.

> The Horizon Table for this project

Each row is one region. Each column tests the hypothesis:

Column	Purpose
region_name	Geographic unit
population	Per-capita denominator
homeless_count	Need (hardest to fill)
funding_eur	Total funding allocated
funding_per_capita	Funding normalised by population
funding_per_homeless	Key column — funding relative to need
source, year	Metadata

> The empty table already tells you where to look

Even without data, the horizon table reveals:

- > **What data we need to find** — population, funding records, homelessness estimates per region
- > **Where the hard parts are** — `homeless_count` is notoriously underreported; whose numbers do we trust?
- > **What analysis we'll do** — compare `funding_per_homeless` across regions
- > **What the output will look like** — a table with 12 rows, one per region

> Who is invisible?

People sleeping rough but not in contact with services.

People in precarious housing who don't count as "homeless" in official statistics.

People in regions where no one is counting at all.

> AI's role in Define

Core rule: the editorial question is always human-driven.

AI can help you:

- > Brainstorm what columns your horizon table might need
- > Suggest what data sources might exist
- > Identify gaps in your table design

AI cannot:

- > Decide what question matters to your community
- > Judge whether a data gap is politically significant
- > Know the context that makes your project necessary

Questions?